

PONCA CITY, OK, USA

WMO: 724530

Lat: 36.737N

Lon: 97.102W

Elev: 1000

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	15.7	-0.6	5.5	15.1	3.8	6.9	19.0	26.2	38.6	23.9	38.0	8.2	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	100.9	74.4	98.3	75.0	95.1	75.0	78.9	92.4	77.8	91.3	76.8	90.0	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	137.4	86.1	73.8	130.5	84.6	72.8	126.3	83.7	43.2	92.3	41.9	91.2	40.8	90.1	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.6	20.3	18.5	DB	2.6	104.4	6.8	3.9	-2.3	107.2	-6.3	109.5	-10.1	111.7	-15.0	114.6	(4)
			WB	1.6	81.5	6.4	2.5	-3.0	83.3	-6.8	84.8	-10.4	86.2	-15.1	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	36.6	40.5	50.0	59.5	68.9	78.3	82.9	81.7	73.6	61.4	49.3	38.7	(6)
(7)		DBStd	18.66	10.76	11.66	11.09	9.18	8.05	5.93	5.75	6.00	8.09	9.39	9.83	10.26	(7)
(8)		HDD50	1414	431	304	137	23	1	0	0	0	17	130	371	(8)	
(9)		HDD65	3807	881	687	476	214	54	1	0	19	180	478	817	(9)	
(10)		CDD50	5145	15	38	138	307	588	850	1020	982	708	371	108	20	(10)
(11)		CDD65	2063	0	1	11	48	176	402	555	517	277	70	6	0	(11)
(12)		CDH74	23923	2	15	114	481	1675	4579	7122	6313	2938	635	49	0	(12)
(13)		CDH80	11674	0	2	19	117	582	2166	3916	3361	1324	182	5	0	(13)
(14)	Wind	WSAvg	9.2	9.0	9.6	10.6	10.8	9.5	9.3	8.4	8.0	8.3	8.8	9.2	8.7	(14)
(15)	Precipitation	PrecAvg	35.6	1.3	1.3	2.4	3.1	4.7	4.6	3.6	3.5	4.0	2.8	2.4	1.5	(15)
(16)		PrecMax	57.6	4.8	4.4	8.0	6.7	10.6	12.2	11.1	11.9	8.2	13.0	12.1	5.4	(16)
(17)		PrecMin	18.7	0.0	0.0	0.0	0.5	0.7	0.5	0.0	0.2	0.1	0.2	0.0	0.0	(17)
(18)		PrecStd	9.1	1.1	1.1	1.9	1.5	2.6	2.4	2.8	2.5	2.2	2.4	2.7	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	70.8	75.9	82.4	88.4	93.3	100.1	105.8	104.9	101.9	89.9	79.3	70.2	(19)	
(20)		MCWB	53.8	57.6	62.1	66.3	71.3	74.9	73.8	74.0	72.9	69.0	62.0	57.5	(20)	
(21)		DB	64.2	70.2	77.1	83.3	89.6	96.5	101.4	100.4	94.7	84.9	74.1	64.3	(21)	
(22)		MCWB	51.1	54.4	59.7	64.1	71.8	74.8	74.6	74.2	73.5	67.6	59.8	53.2	(22)	
(23)		DB	59.2	64.5	72.7	79.8	86.2	93.3	98.6	97.6	91.2	81.1	70.1	59.3	(23)	
(24)		MCWB	48.7	52.5	58.3	62.7	70.4	74.4	75.7	74.6	72.6	65.5	58.3	50.2	(24)	
(25)		DB	54.0	59.6	68.3	75.7	82.8	90.6	95.3	94.3	87.8	76.9	65.7	55.0	(25)	
(26)		MCWB	44.8	49.0	56.1	61.7	69.3	74.0	75.8	74.8	71.4	64.3	55.4	47.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	59.9	61.4	66.5	70.3	76.4	80.1	80.9	80.2	77.7	73.4	66.7	62.1	(27)	
(28)		MCDWB	65.2	68.3	76.7	80.7	87.4	93.4	93.6	93.1	92.1	83.5	73.8	67.5	(28)	
(29)		WB	53.5	58.1	63.8	68.1	74.3	78.1	79.1	78.3	75.6	71.3	62.9	56.4	(29)	
(30)		MCDWB	61.6	66.5	72.0	78.0	85.2	91.0	92.5	91.7	89.0	79.9	70.1	61.6	(30)	
(31)		WB	48.9	54.1	61.0	66.3	72.7	76.8	78.0	77.2	74.3	69.0	60.2	51.5	(31)	
(32)		MCDWB	58.2	63.3	69.7	76.1	82.9	89.0	91.8	90.5	87.0	76.9	67.3	58.0	(32)	
(33)		WB	45.2	49.5	57.7	63.9	71.0	75.3	77.0	76.0	73.0	66.4	57.3	47.1	(33)	
(34)		MCDWB	53.1	59.2	67.4	73.3	80.5	86.8	90.7	89.1	84.6	74.5	64.8	54.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	21.3	22.3	23.0	22.7	20.6	20.0	20.6	20.9	21.7	22.5	22.3	20.1	(35)	
(36)		5% DB	29.9	29.8	28.6	26.8	23.1	22.3	23.9	24.1	23.6	25.6	27.2	27.0	(36)	
(37)		MCWBR	18.4	18.0	15.8	14.1	10.1	8.0	6.7	7.2	8.7	12.1	15.7	17.5	(37)	
(38)		5% WB	26.0	26.0	23.2	22.5	19.8	19.5	20.6	20.2	20.1	19.5	22.1	23.2	(38)	
(39)		MCWBR	17.8	17.7	14.9	13.9	10.6	9.0	7.9	7.6	8.3	11.0	15.6	17.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.301	0.340	0.381	0.420	0.428	0.435	0.436	0.392	0.345	0.306	0.285	(40)	
(41)		taud	2.497	2.485	2.401	2.309	2.272	2.285	2.303	2.245	2.351	2.471	2.512	2.527	(41)	
(42)		Ebn at Noon	288	298	294	285	275	272	269	266	273	278	280	284	(42)	
(43)		Edh at Noon	25	29	35	41	43	42	41	43	36	30	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	832	1044	1361	1699	1885	2150	2133	1922	1604	1198	889	704	(44)	
(45)		RadStd	82	154	89	115	181	174	123	115	125	142	105	71	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page